2019-Ph-H1-Q7
Section: Our Dynamic Universe
Topic: Collisions, Explosions and Impulse

Question:
A ball of mass 0.2 kg strikes a wall and rebounds. Its velocity changes from +4 m/s to −2 m/s in 0.01 s. What is the average force?

Answer:
Impulse = Δp = $m(v_{\text{final}} - v_{\text{initial}})$
= $0.2 \times (-2 - 4) = 0.2 \times (-6) = -1.2\,\text{kg}\,\text{m/s}$
Force = impulse / time = $-1.2/0.01 = -120\,\text{N}$ (negative = direction change)

Correct Option: C (120 N)

Guidance for Students:
Use impulse-momentum theorem: force × time = change in momentum. Don’t forget to subtract vectors correctly.

Revision Tips:

- Impulse = change in momentum = $F\Delta t$
- Watch the **signs** in velocity change
- Time must be in **seconds**